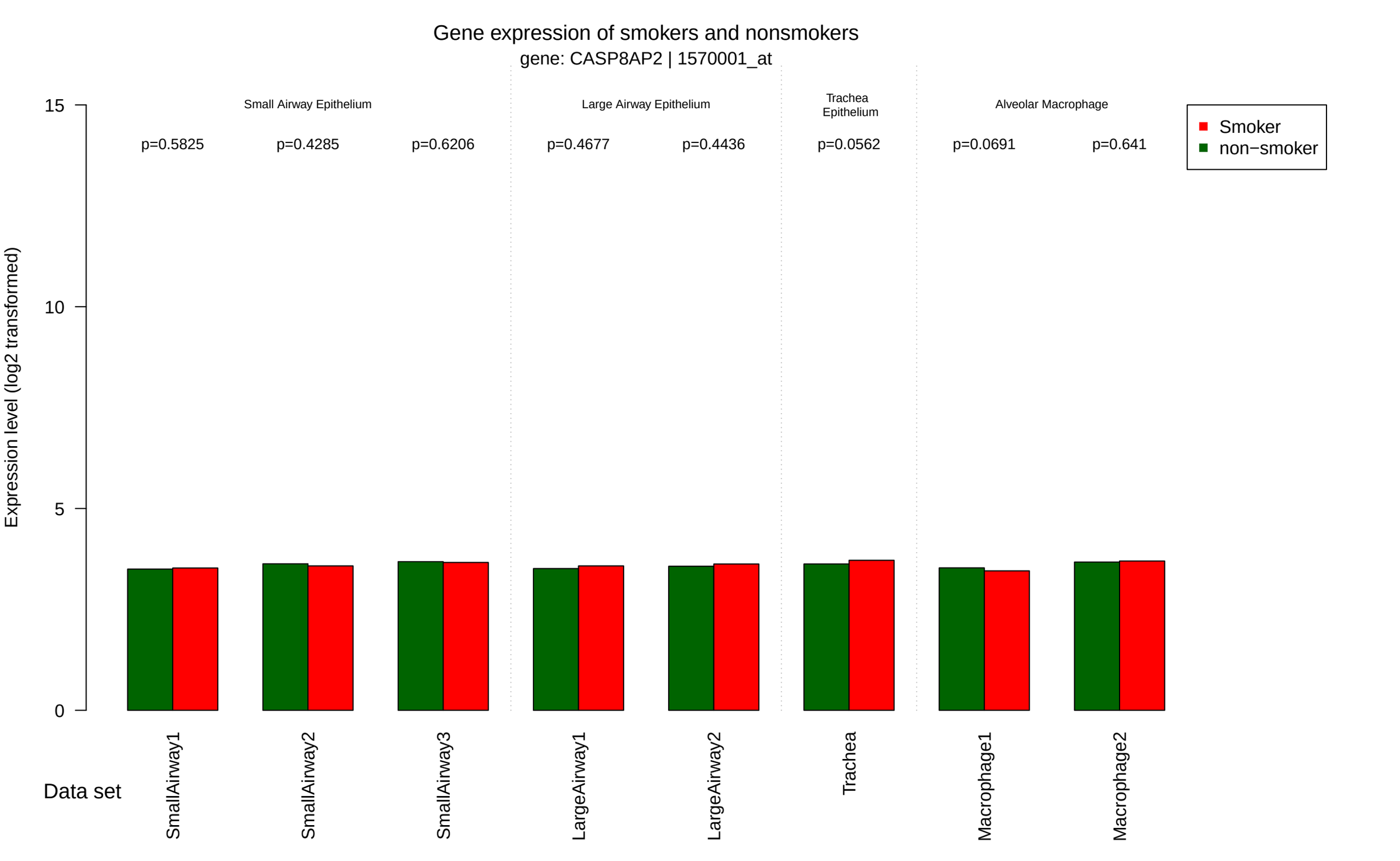
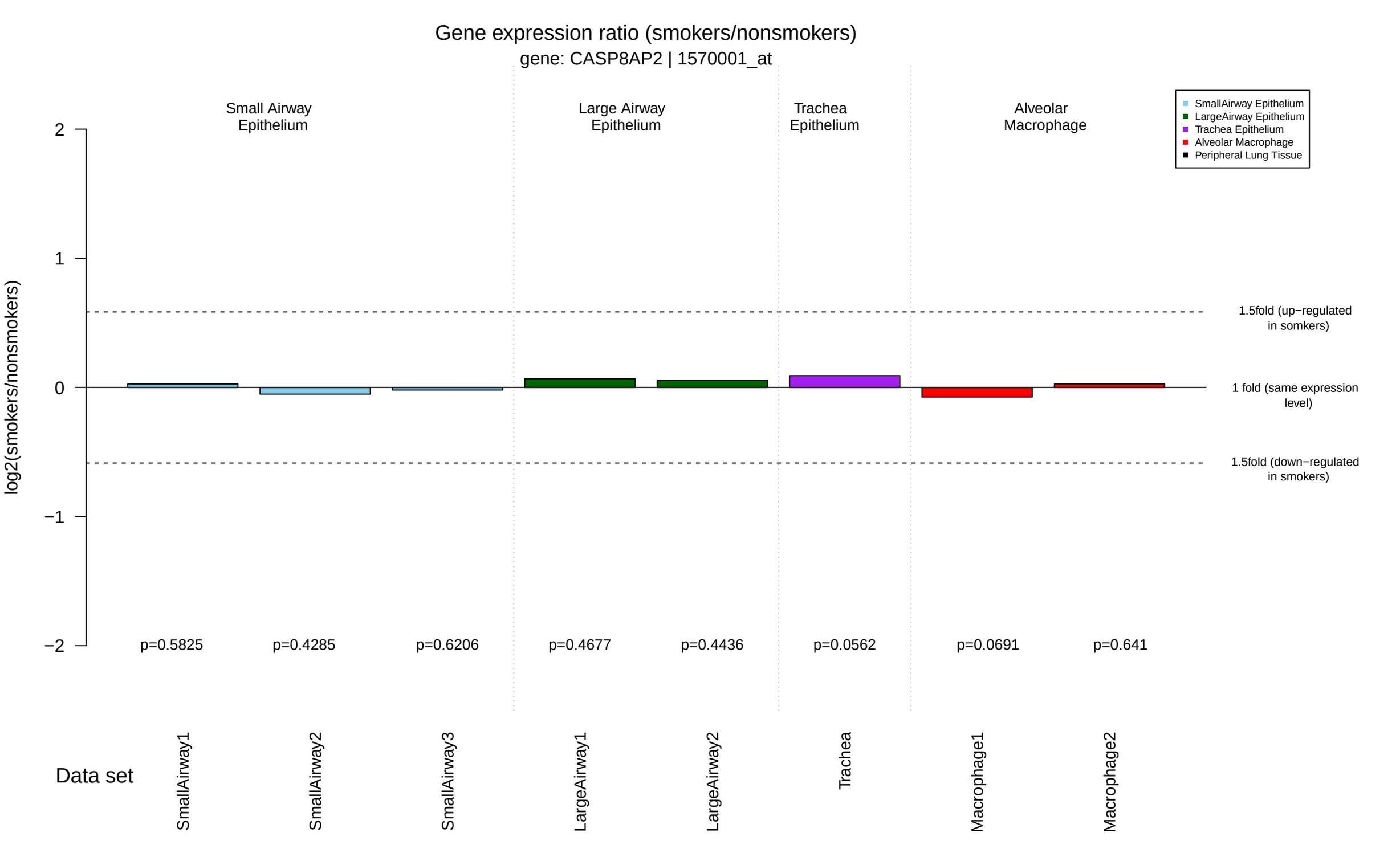
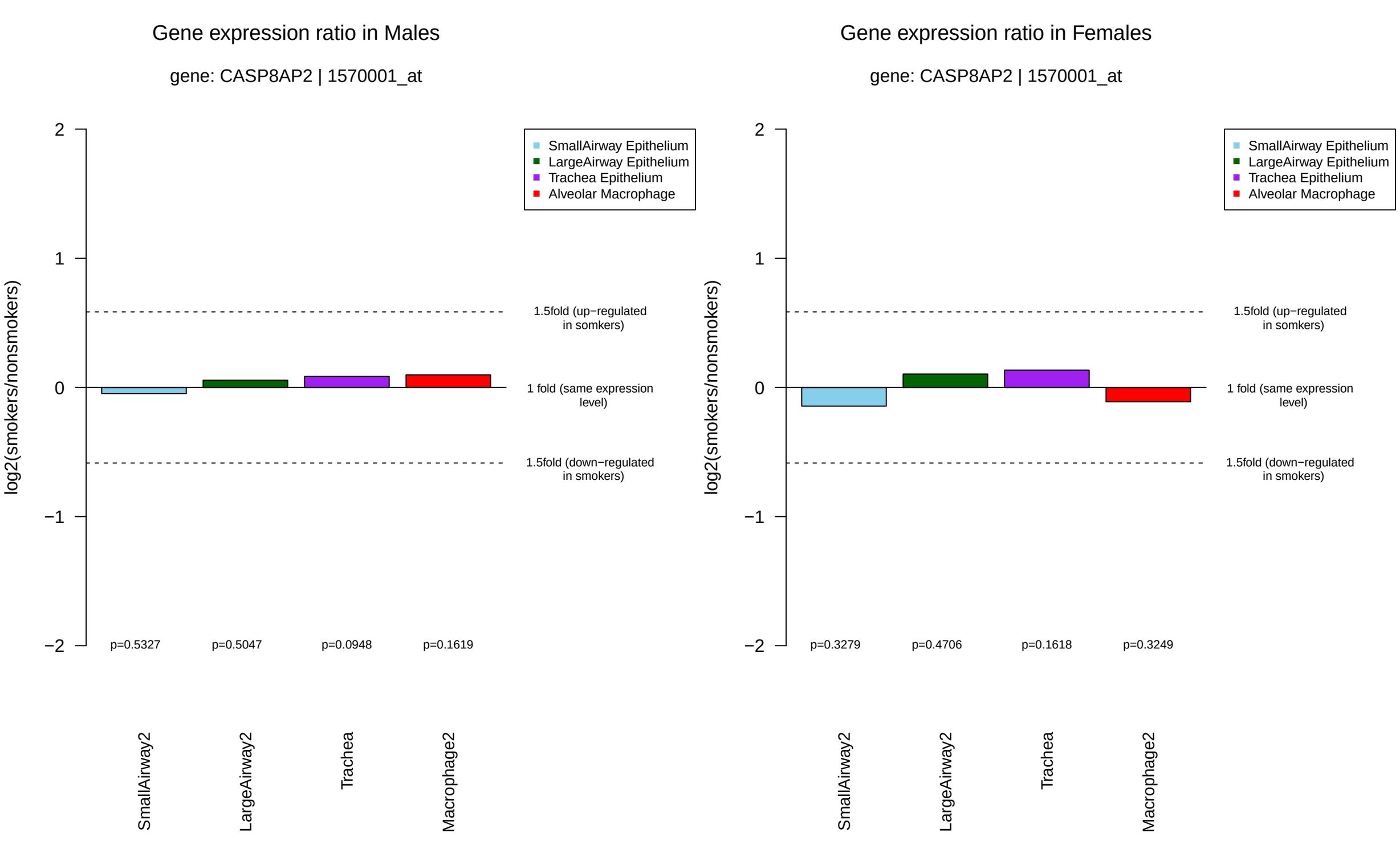
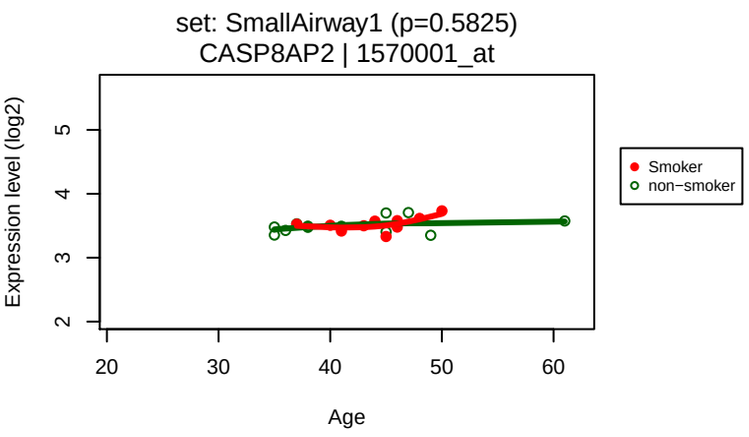
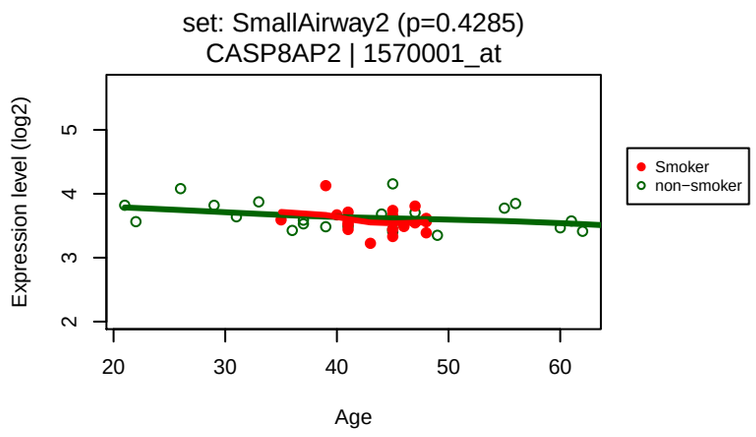
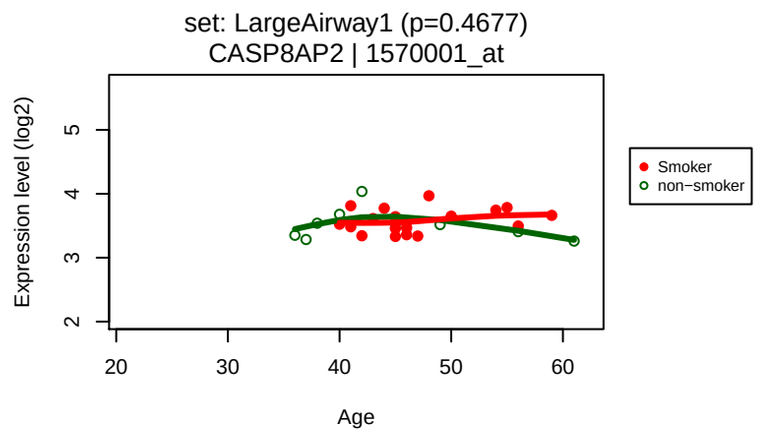

Gene expression of smokers and nonsmokers
gene: CASP8AP2 | 1570001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5825
p=0.4285
p=0.6206
p=0.4677
p=0.4436
p=0.0562
p=0.0691
p=0.641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

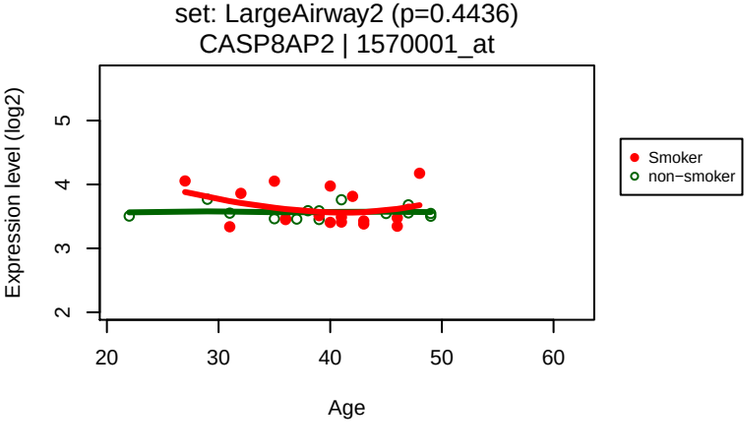

Gene expression ratio (smokers/nonsmokers)
gene: CASP8AP2 | 1570001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5825
p=0.4285
p=0.6206
p=0.4677
p=0.4436
p=0.0562
p=0.0691
p=0.641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

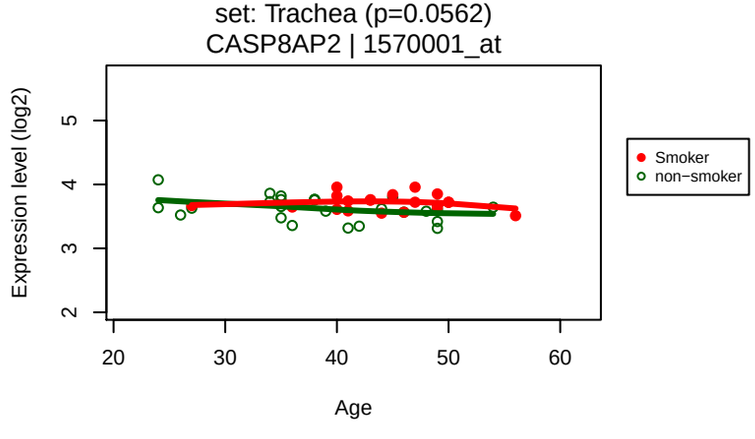

Gene expression ratio in Males
gene: CASP8AP2 | 1570001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5327
p=0.5047
p=0.0948
p=0.1619
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASP8AP2 | 1570001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3279
p=0.4706
p=0.1618
p=0.3249
SmallAirway2
LargeAirway2
Trachea
Macrophage2

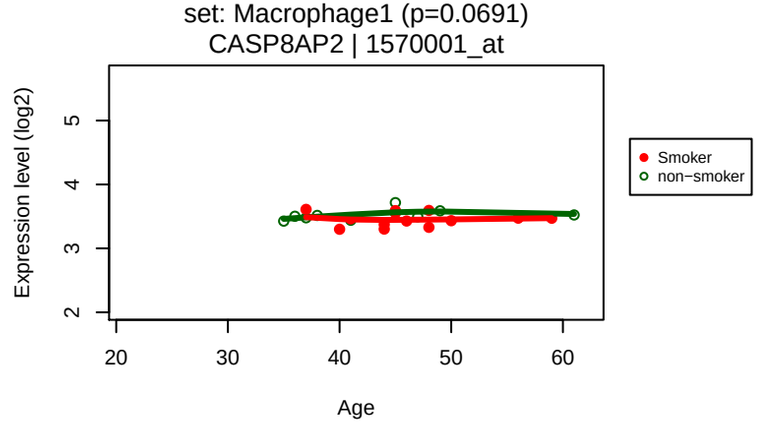

set: SmallAirway1 (p=0.5825)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker

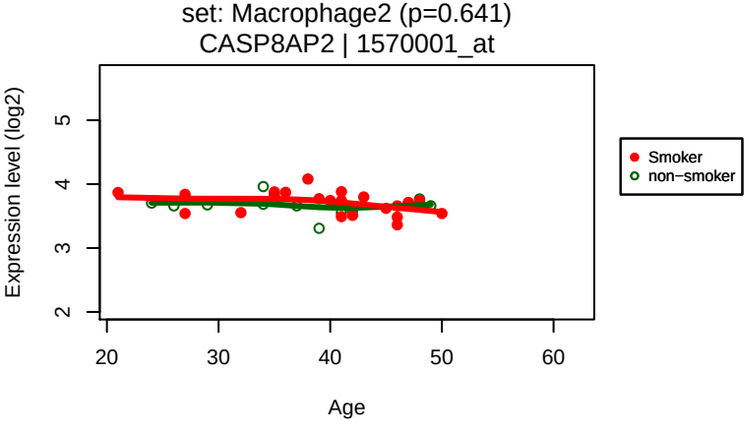

set: SmallAirway2 (p=0.4285)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker

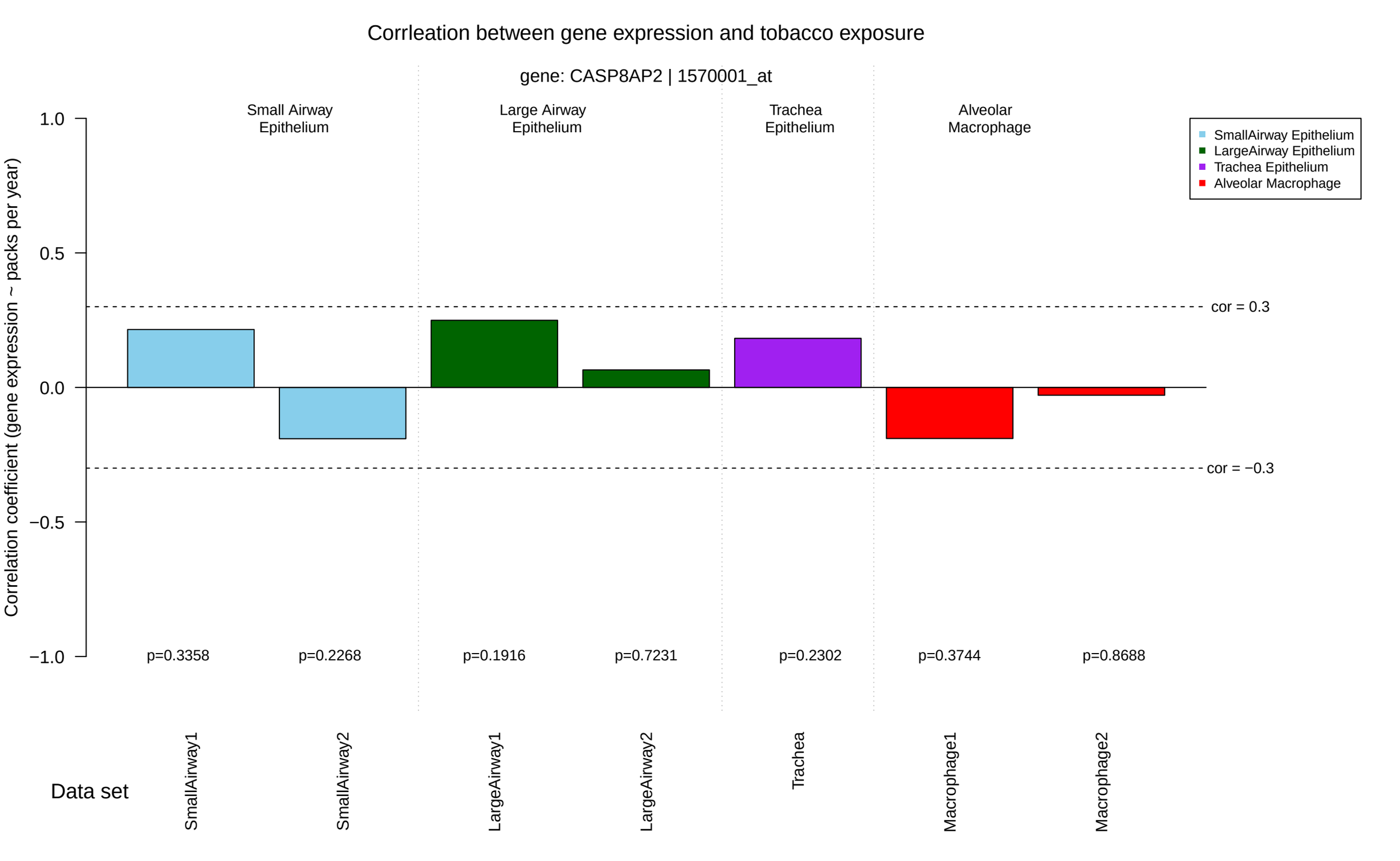

set: LargeAirway1 (p=0.4677)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4436)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0562)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0691)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.641)
CASP8AP2 | 1570001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASP8AP2 | 1570001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3358
p=0.2268
p=0.1916
p=0.7231
p=0.2302
p=0.3744
p=0.8688
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2